

Data Cluster Security & Edge Device Protocols

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1. Purpose & Domain Definition

B-Cubed distributed applications often process operational and educational management telemetry at the network edge. This policy dictates security requirements governing local hardware clusters, testbeds, container environments, and credentials lifecycle management.

2. Distributed Edge & Micro-Cluster Regulations

When engineering physical testbeds, local network nodes, or proof-of-concept AI frameworks deployed across local bare-metal components (such as multi-node ARM-based single-board computers), the following constraints apply:

2.1 Network Isolation and Boundary Security

- **Dedicated Subnets:** All hardware arrays must be locked to isolated, unrouted staging virtual networks. Traffic crossing outward segments must utilize encrypted TLS transport loops.
- **Physical Asset Registry:** Edge clusters must have assigned physical identity tracking tags, stating the designated node inventory counts and localized hardware models.

System Load Balancing Formula

The total throughput metric allocation across distributed local configurations is scaled according to operational density rules:

$$C_{capacity} = \Sigma (N_{nodes} \times \mu_{proc}) \times (1 - \Delta_{overhead})$$

Where total cluster capacity requires systematic balancing adjustments to safely prevent single-node compute exhaustion.

2.2 Access Control and Secure Configuration

Default factory configuration passwords must be replaced instantly with unique cryptographic credentials generated via organizational vaults. Root level shell access over Secure Shell (SSH) is strictly forbidden; keys must map to named unprivileged developer credentials profiles.

3. Authentication & Identity Management Architecture

User session validation, backend identity workflows, and consumer data partitions must rely entirely on authorized authentication providers. B-Cubed engineering explicitly mandates decoupled identity-as-a-service frameworks.

Authentication State Trigger	Requirement Parameter
User Token Expiry	Maximum TTL of 3600 seconds; refresh configurations require strict cryptographically signed client contexts.
Multi-Factor Verification	Mandatory enforcement across administrative, engineering, and support control planes.
API Communication Security	Bearer Token structure utilizing verified external platform sign-offs; secret keys rotation occurs every 90 days.

4. Regulatory Enforcement and Incident Control

Non-compliance with information governance patterns poses critical compliance and infrastructure vulnerability risk. Periodic automated monitoring routines run continuously across active repositories to isolate leaked application secrets or unencrypted network configurations.